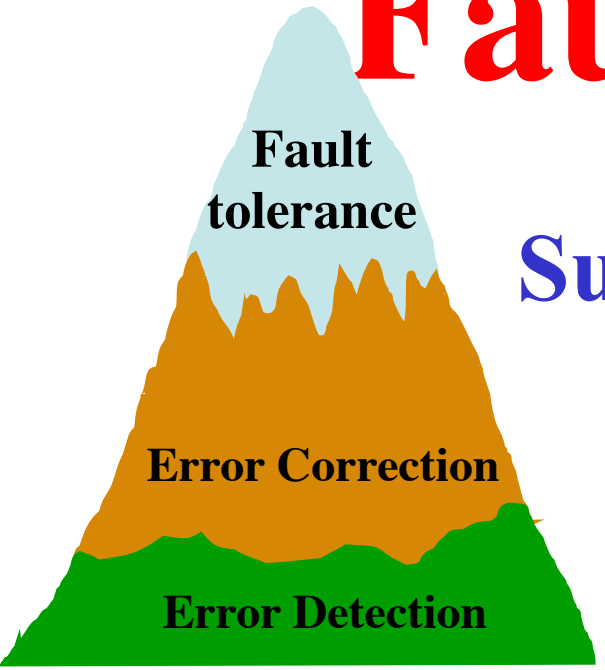
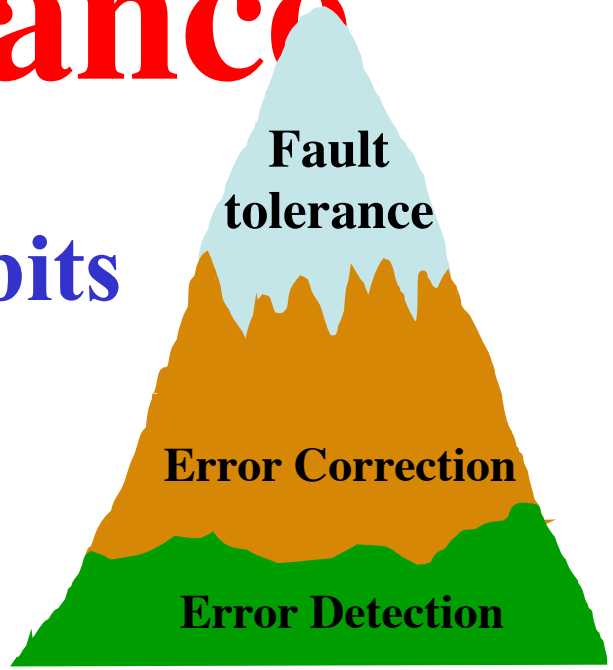
Climbing Mount Fault-Tolerance

A Trailhead: Supercoherent Qubits





Dave Bacon

Institute for Quantum Information

California Institute of Technology

Computer Revolution = Physics

The Computer Science Assumption: *robust* digital storage and digital manipulation of information is possible.

In this NOISY (ANALOG) world HOW can anything be DIGITAL?

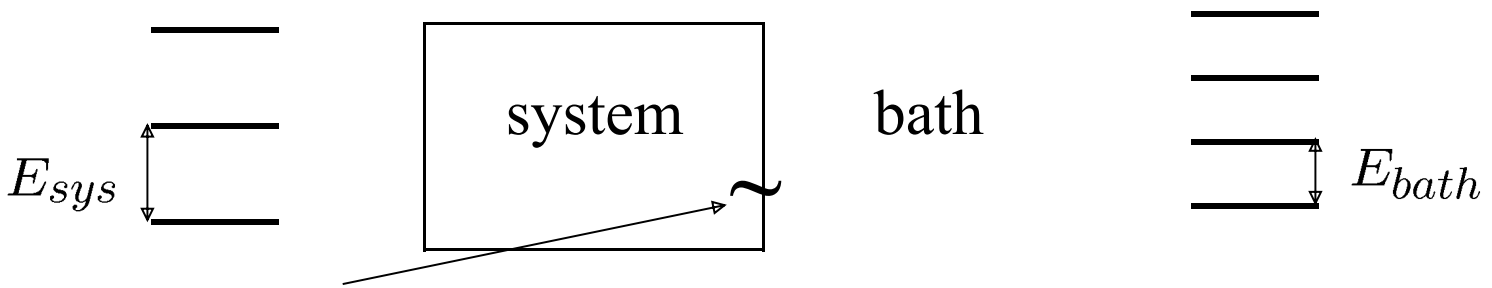
Theory: Von Neumann showed methods for fault-tolerant computation

Practice: no need for these methods
because
THERE EXIST SYSTEMS WHOSE PHYSICS GUARANTEES ROBUST CLASSICAL COMPUTATION

THE BILLION DOLLAR QUANTUM QUESTION

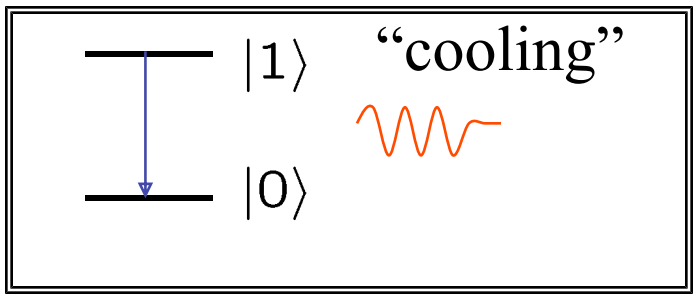
Do there exist (or can we engineer) quantum systems whose physics guarantees fault tolerant quantum computation?

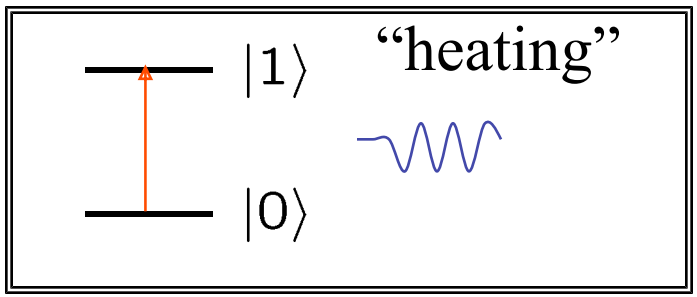
Making Decoherence Work Hard

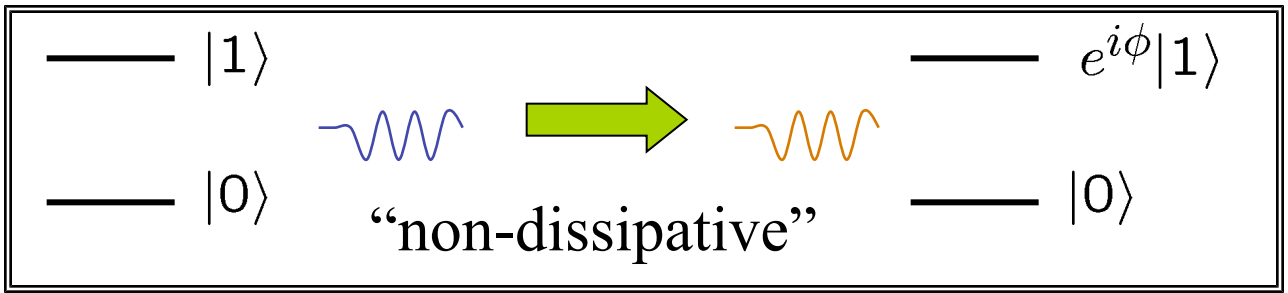


HI = perturbative system-bath coupling

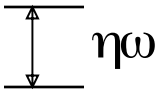
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies*
(essentially rotating wave approximation)







two-level atom radiatively coupled to a thermal reservoir:



$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right]$$

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right]$$

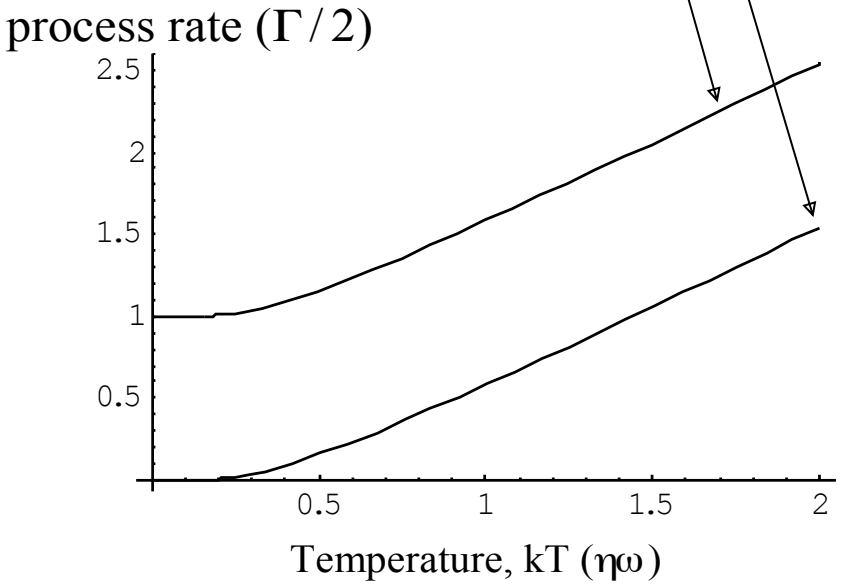
$$-\phi\left[\rho - \sigma_z\rho\sigma_z\right]$$

"heating"

"cooling"

"non-dissipative"

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right)-1}$$



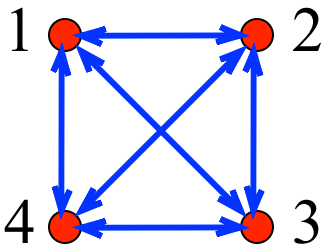
At low bath temperatures, heating disappears……..

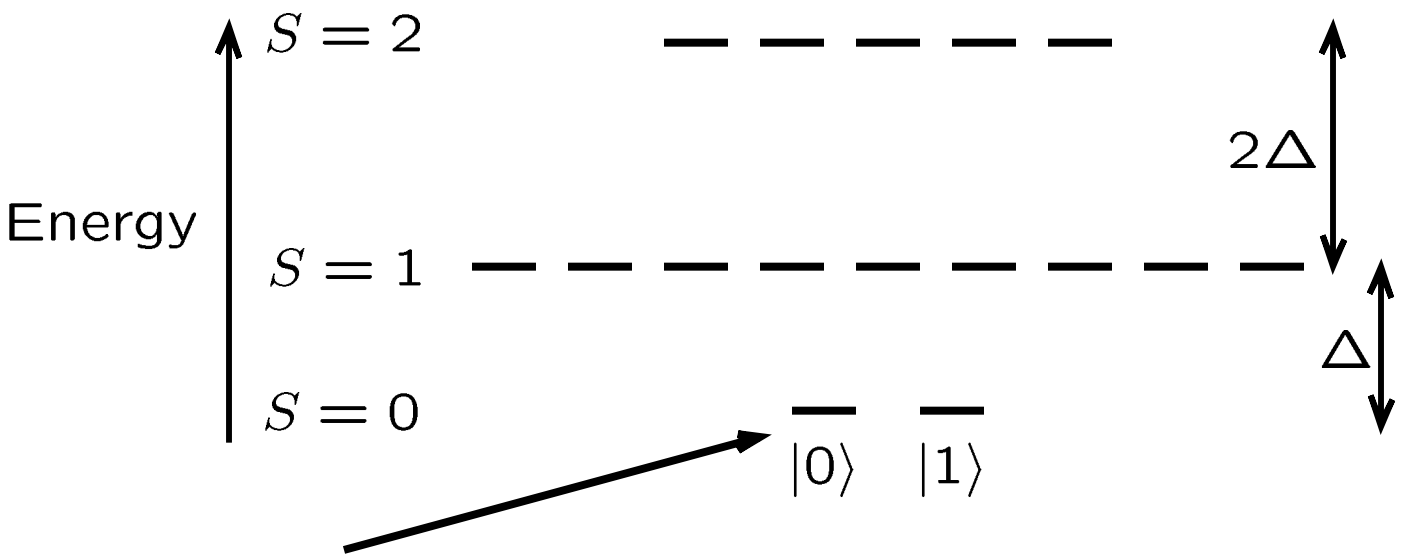
Can we make quantum errors "heating"?

Got Exchange?

Four spin systems with equal strength exchange interactions between all spins:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\boldsymbol{s}}_j = \frac{\Delta}{2}\hat{\mathbf{S}}^2 + c\mathbf{I}$$



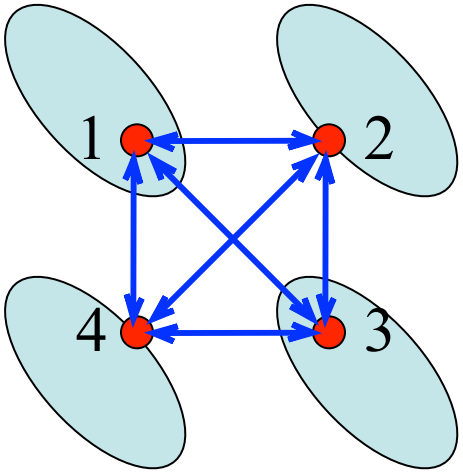


Encode a qubit of quantum information in the degenerate ground state

Stabilized by collective operators: $\mathbf{GGGG}|\psi\rangle = |\psi\rangle$

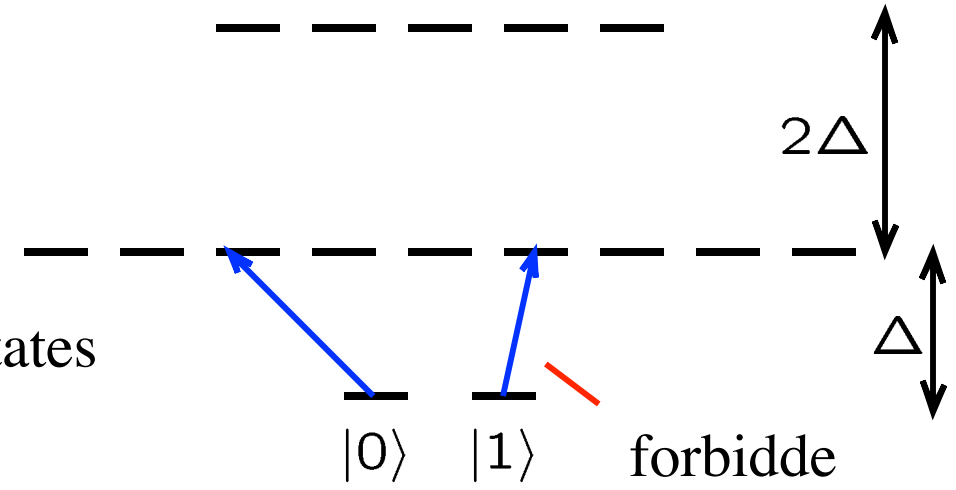
Degenerate ground states are single qubit error detecting code

Supercoherence



Qubits coupling to baths:

$$\mathbf{H}_I = \sum_{i=1}^{4} \hat{\mathbf{s}}_i \cdot \hat{\mathbf{B}}_i$$



All single qubit processes on this system take the ground states to states of higher energy.

As temperature of the bath is decreased $T \ll \Delta$ first-order decoherence processes will be eliminated…supercoherence

Limit to supercoherence is second order processes which act on two qubits

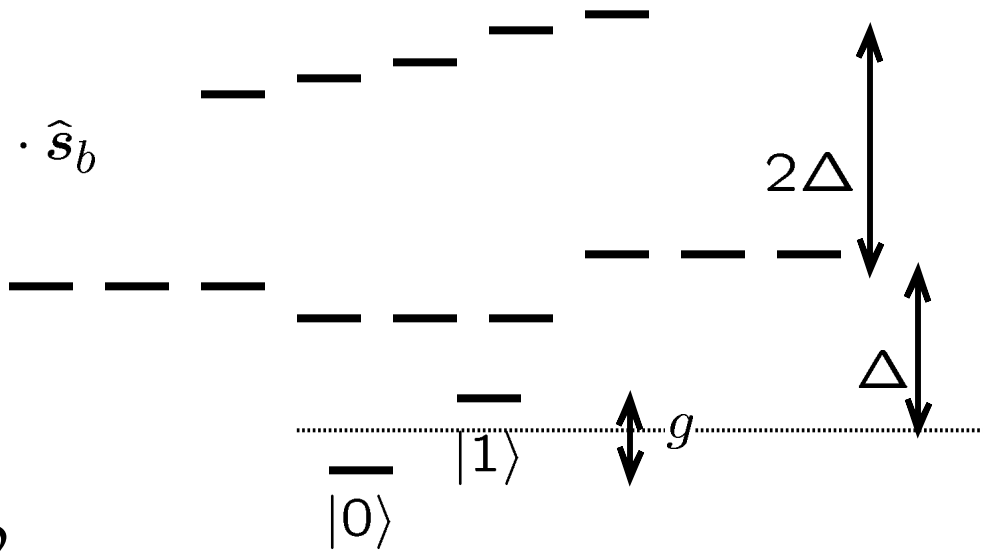
Bacon, Brown, and Whaley, **PRL** 87, 247902 (2001)

Computing on Supercoherence

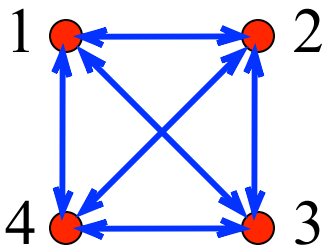
Manipulation of the encoded qubit can be enacted by additional exchange energy:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\boldsymbol{s}}_j + g\hat{\boldsymbol{s}}_a \cdot \hat{\boldsymbol{s}}_b$$

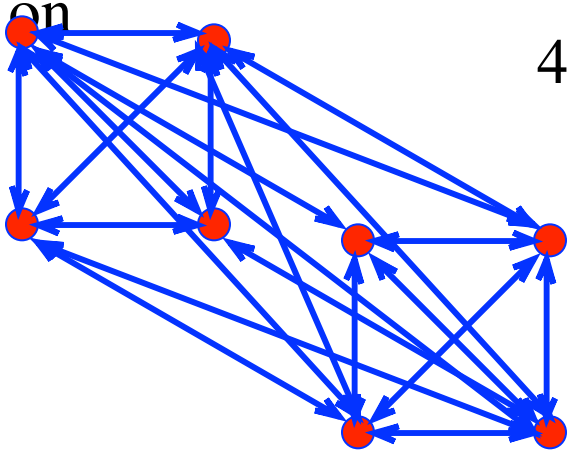
$g \ll T \Rightarrow$ supercoherence property not lost.



Extra exchange on



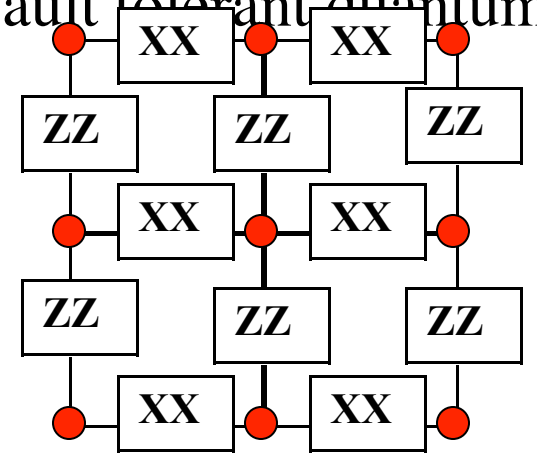
gives all SU(2) on encoded qubit



To compute on between two encoded qubits exchange must be turned on between eight physical qubits

Beyond Supercoherence

Supercoherence is small step towards the ultimate goal of natural
fault tolerant quantum computation.

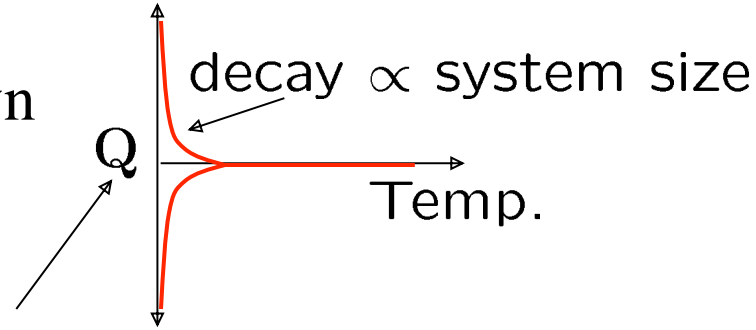


This systems ground state is a quantum error correcting code.

Kitaev and Freedman's topological codes are examples of systems whose energy eigenstates are quantum error correcting code states.

Problem: find a system with a persistent coherence order parameter

All known systems



The goal

